

Self-Check Questions: Dedekind Cuts

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1 Definitions of a Dedekind Cut

1. State the two-set definition of a Dedekind cut (A, B) of $\mathbb{Q}$.

Answer. A Dedekind cut is a pair (A, B) of nonempty subsets of $\mathbb{Q}$ such that

$$A \cup B = \mathbb{Q}, \quad A \cap B = \emptyset,$$

every $a \in A$ is less than every $b \in B$, and A has no greatest element.

2. State the one-set definition using only the lower set $A \subseteq \mathbb{Q}$.

Answer. A Dedekind cut may be defined as a nonempty proper subset $A \subsetneq \mathbb{Q}$ that is downward closed and has no greatest element.

3. Why is it enough to specify only A ? How is B recovered?

Answer. Because A and B partition $\mathbb{Q}$,

$$B = \mathbb{Q} \setminus A.$$

Thus either half uniquely determines the other.

4. What does it mean for A to be downward closed?

Answer. Downward closed means

$$x \in A, y < x \implies y \in A.$$

5. Write the statement “ A has no greatest element” entirely with quantifiers.

Answer.

$$\forall a \in A \exists a' \in A \text{ such that } a < a'.$$

6. Are the following two statements equivalent?

A has no greatest element

and

$$\forall a \in A \exists a' \in A \text{ such that } a < a'.$$

Explain.

Answer. Yes. If every $a \in A$ has a strictly larger element of A , then no element can be greatest, and conversely.

7. Why must a Dedekind lower set satisfy both

$$A \neq \emptyset \quad \text{and} \quad A \neq \mathbb{Q}?$$

Answer. $A \neq \emptyset$ prevents the trivial empty lower half, while $A \neq \mathbb{Q}$ ensures that there is a nonempty upper half.

8. Suppose A is nonempty, proper, and downward closed, but is allowed to have a greatest element. What extra objects would then be admitted?

Answer. Such sets would include versions of rational cuts whose lower half contains its endpoint, for example

$$\{q \in \mathbb{Q} : q \leq 3\}.$$

This creates two possible cuts for the same rational boundary unless a convention is imposed.

9. Why is the “no greatest element” condition partly a convention about where a rational boundary is placed?

Answer. For a rational boundary r , one must decide whether r belongs to the lower or upper half. Requiring A to have no greatest element forces r into B .

10. Give the cut corresponding to the rational number 3 using the usual lower-set convention.

Answer.

$$A = \{q \in \mathbb{Q} : q < 3\}, \quad B = \{q \in \mathbb{Q} : q \geq 3\}.$$

11. Could one instead define rational cuts using

$$A = \{q \in \mathbb{Q} : q \leq r\}?$$

What would have to change in the definition?

Answer. Yes, but then the axioms would need to be changed, for example by requiring B to have no least element instead. It is a different endpoint convention.

12. State the order-theoretic definition of a Dedekind cut using the phrase “initial segment.”

Answer. A Dedekind cut is a nonempty proper initial segment of $(\mathbb{Q}, <)$ with no greatest element.

2 Maximum, Upper Bound, and Supremum

1. Define what it means for m to be the maximum of a set S .

Answer. m is the maximum of S if $m \in S$ and

$$x \leq m$$

for every $x \in S$.

2. Define what it means for u to be an upper bound of S .

Answer. u is an upper bound of S if

$$x \leq u$$

for every $x \in S$. It need not belong to S .

3. Define what it means for s to be the supremum of S .

Answer. s is the supremum of S if s is an upper bound and every upper bound u satisfies

$$s \leq u.$$

4. What is the logical difference between

$$\max S \quad \text{and} \quad \sup S?$$

Answer. A maximum must belong to S . A supremum need not belong to S .

5. Can a set have a supremum without having a maximum? Give an example in $\mathbb{Q}$.

Answer. Yes. For example

$$S = \{q \in \mathbb{Q} : q < 1\}$$

has supremum 1 in $\mathbb{Q}$ but no maximum.

6. Can a set have a maximum without having a supremum?

Answer. No. If $m = \max S$, then m is an upper bound of S , and since $m \in S$, every upper bound u of S satisfies $m \leq u$. Hence m is the least upper bound, so

$$\sup S = m.$$

This holds in any ordered set, with no completeness assumption needed.

7. If A is the lower half of a Dedekind cut, does A have an upper bound in $\mathbb{Q}$? Prove your answer directly from the cut axioms.

Answer. Yes. Choose any $b \in B$. Since every $a \in A$ satisfies $a < b$, the number b is a rational upper bound of A .

8. If $A \subseteq \mathbb{Q}$ is bounded above, must $\sup_{\mathbb{Q}} A$ exist?

Answer. No. For example,

$$\{q \in \mathbb{Q} : q^2 < 2, q > 0\}$$

is bounded above in $\mathbb{Q}$ but has no rational supremum.

9. Why is the notation

$$\sup A$$

potentially ambiguous in a foundational discussion?

Answer. The ambient ordered set is omitted. A set may have a supremum in $\mathbb{R}$ but not in $\mathbb{Q}$.

10. Explain the difference between

$$\sup_{\mathbb{Q}} A \quad \text{and} \quad \sup_{\mathbb{R}} A.$$

Answer. $\sup_{\mathbb{Q}} A$ must itself be rational, while $\sup_{\mathbb{R}} A$ may be irrational.

11. Consider

$$A = \{q \in \mathbb{Q} : q < 0 \text{ or } q^2 < 2\}.$$

Does A have a supremum in $\mathbb{Q}$?

Answer. No. Its least upper bound would have to be $\sqrt{2}$, which is not rational.

12. If $\mathbb{R}$ has already been constructed, what is $\sup_{\mathbb{R}} A$ for the set above?

Answer.

$$\sup_{\mathbb{R}} A = \sqrt{2}.$$

13. Why would using $\sup_{\mathbb{R}} A$ during the construction of $\mathbb{R}$ risk circular reasoning?

Answer. If $\mathbb{R}$ is being constructed from cuts, using completeness of $\mathbb{R}$ to define or justify the cuts assumes the object being constructed.

14. Does “ A has no greatest element” imply “ A has no supremum”?

Answer. No. For example,

$$A = \{q \in \mathbb{Q} : q < 1\}$$

has no greatest element but has supremum 1 in $\mathbb{Q}$.

15. Does “ A has no greatest element” imply that A is unbounded above?

Answer. No. The same set is bounded above by 1.

3 What Can Be Proved Using Only $\mathbb{Q}$?

1. Before $\mathbb{R}$ is constructed, which number system are the elements of a Dedekind cut taken from?

Answer. Every element of a cut is rational. The construction begins entirely inside $\mathbb{Q}$.

2. Which of the following can be proved using only the cut axioms and the order properties of $\mathbb{Q}$?

- (a) A is bounded above.
- (b) A has a rational supremum.
- (c) A has no greatest element.
- (d) B is upward closed.
- (e) Every element of B is an upper bound of A .

Answer. (a), (c), (d), and (e) can be proved. Statement (b) is false in general because $\mathbb{Q}$ is not complete.

3. Why is it legitimate to use density of $\mathbb{Q}$ when constructing $\mathbb{R}$ from cuts?

Answer. Density of $\mathbb{Q}$ is already a theorem about rational numbers, so it does not depend on the existence of $\mathbb{R}$.

4. Why is it not legitimate to use completeness of $\mathbb{R}$ at that stage?

Answer. Completeness of $\mathbb{R}$ is one of the properties the construction is intended to establish, so using it beforehand would be circular.

5. Does the Archimedean property belong only to $\mathbb{R}$?

Answer. No. An Archimedean property can already be proved for $\mathbb{Q}$.

6. State an Archimedean property that can be proved already for $\mathbb{Q}$.

Answer. One version is: for every $x \in \mathbb{Q}$ there exists $n \in \mathbb{N}$ such that

$$n > x.$$

7. If a proof about cuts invokes the Archimedean property, what must you check to make sure the proof is not circular?

Answer. You must check that the version used was proved from properties of $\mathbb{Q}$ or $\mathbb{N}$, rather than imported from completeness of $\mathbb{R}$.

8. Suppose someone says: “Since A is bounded above, it has a least upper bound.” What hidden assumption has been made?

Answer. The hidden assumption is completeness of the ambient ordered set.

9. Suppose someone says: “The cut has a boundary point c , and therefore $A = \{q \in \mathbb{Q} : q < c\}$.” What has potentially been assumed without proof?

Answer. The argument assumes that such a boundary object c already exists in some larger ordered set.

10. Give an example of a statement about cuts that is true after $\mathbb{R}$ is constructed but cannot be used as a premise in constructing $\mathbb{R}$.

Answer. For example: “Every nonempty bounded subset of $\mathbb{R}$ has a supremum.” This is valid only after $\mathbb{R}$ and its completeness have been established.

4 Rational Cuts and Irrational Cuts

1. For $r \in \mathbb{Q}$, define the lower set

$$A_r = \{q \in \mathbb{Q} : q < r\}.$$

Answer.

$$A_r = \{q \in \mathbb{Q} : q < r\}.$$

2. What is the corresponding upper set B_r ?

Answer.

$$B_r = \{q \in \mathbb{Q} : q \geq r\}.$$

3. Does A_r have a greatest element?

Answer. No. Between any $q < r$ and r there is another rational.

4. Does B_r have a least element?

Answer. Yes. Its least element is r .

5. Prove that if a cut is produced by a rational number r under the usual convention, then B has a least element.

Answer. By definition,

$$B_r = \{q \in \mathbb{Q} : q \geq r\},$$

so $r \in B_r$ and $r \leq b$ for every $b \in B_r$.

6. Suppose B has a least element b . Show that the cut is exactly the rational cut determined by b .

Answer. If $b = \min B$, then every rational $q < b$ cannot lie in B , so $q \in A$. Also no rational $q \geq b$ lies in A . Hence

$$A = \{q : q < b\}.$$

7. Deduce the intrinsic characterization

the cut comes from a rational number $\iff B$ has a least element.

Answer. The two implications above prove

$$B \text{ has a least element} \iff \text{the cut is induced by some } r \in \mathbb{Q}.$$

8. How would you characterize a “new” cut, one not corresponding to any rational number, using only the internal structure of B ?

Answer. It is a cut whose upper set B has no least element.

9. Why is this characterization preferable, during the construction, to saying “the boundary is irrational”?

Answer. Because “irrational boundary” already presupposes the existence of irrational real numbers. The property of B is internal to $\mathbb{Q}$.

10. For the cut associated with $\sqrt{2}$, why is there no least rational element of B ?

Answer. If $b \in B$, then $b > \sqrt{2}$. By density, there exists rational b' with

$$\sqrt{2} < b' < b,$$

so b cannot be least.

5 The Dangerous Word “Boundary”

1. What does the expression

$$A = \{q \in \mathbb{Q} : q < c\}$$

presuppose about c ?

Answer. It presupposes that c is already an element of some ordered set in which comparison with rationals is meaningful.

2. If $c \in \mathbb{Q}$, is there any foundational problem with this definition?

Answer. No. If $c \in \mathbb{Q}$, the definition uses only rational order.

3. If $c \notin \mathbb{Q}$ and $\mathbb{R}$ has not yet been constructed, what exactly is c supposed to be?

Answer. Without an already constructed extension of $\mathbb{Q}$, such a c has not yet been defined.

4. Why can the geometric picture of a cut as “all rationals to the left of a boundary” be conceptually useful but logically dangerous?

Answer. The picture suggests a pre-existing endpoint, whereas in the construction the cut itself is the new number.

5. In the Dedekind construction, is the real number first given and then encoded by a cut, or is the cut itself taken to be the real number?

Answer. The cut itself is taken to be the real number.

6. Rewrite the statement “this cut has boundary $\sqrt{2}$ ” in a way that does not presuppose that $\sqrt{2}$ already exists as a real number.

Answer. One can say: define the object called $\sqrt{2}$ to be the cut

$$\{q \in \mathbb{Q} : q < 0 \text{ or } q^2 < 2\}.$$

7. Suppose a cut has no rational boundary. Does that mean it has no meaning or no location in the order?

Answer. No. The cut itself specifies a position in the ordering even when no rational lies exactly at that position.

8. What does it mean to say that a cut identifies a “gap” in $\mathbb{Q}$?

Answer. It means the partition behaves like a missing order-location in $\mathbb{Q}$: all rationals on one side lie below all rationals on the other, but no rational realizes the boundary.

6 How an Irrational Number Is Defined Without Naming It First

1. Consider

$$A = \{q \in \mathbb{Q} : q < 0 \text{ or } q^2 < 2\}.$$

Why can this set be defined without assuming the existence of the real number $\sqrt{2}$?

Answer. The defining condition uses only rational inequalities and rational squaring. No prior real number is required.

2. Verify that this set is nonempty.

Answer. For example, $0 \in A$.

3. Verify that this set is a proper subset of $\mathbb{Q}$.

Answer. For example, $2 \notin A$, so $A \neq \mathbb{Q}$.

4. Prove that it is downward closed.

Answer. If $x \in A$ and $y < x$, then either $x < 0$, which gives $y < 0$, or $x \geq 0$ and $x^2 < 2$. In the latter case, negative y is automatically in A , while $0 \leq y < x$ gives $y^2 < x^2 < 2$.

5. Prove that it has no greatest element.

Answer. For negative q , choose a slightly larger negative rational. For nonnegative q with $q^2 < 2$, one can explicitly choose a sufficiently small positive rational increment that keeps the square below 2.

6. Which part of that proof requires the most care?

Answer. The last step: constructing a larger rational while preserving $q^2 < 2$ without using $\sqrt{2}$.

7. Once this set is proved to be a Dedekind cut, what object is taken to be $\sqrt{2}$ in the Dedekind construction?

Answer. The cut itself is defined to be the real number $\sqrt{2}$.

8. Does every irrational real number need to have a simple algebraic description like $q^2 < 2$?

Answer. No. Most irrational numbers do not have a simple algebraic formula.

9. If not, how can the construction still contain all real numbers?

Answer. The collection of all Dedekind cuts is taken as the set of real numbers. A cut need not have a simple finite description to exist as a set.

10. What is conceptually wrong with asking: “How do we define the cut belonging to an arbitrary irrational number before irrational numbers have been defined?”

Answer. It presupposes that the irrational number exists independently before the construction. The direction is reversed: the cut defines the number.

11. After $\mathbb{R}$ has been constructed, how can one show that every real number corresponds to a Dedekind cut of $\mathbb{Q}$?

Answer. Given $\alpha \in \mathbb{R}$, define

$$A_\alpha = \{q \in \mathbb{Q} : q < \alpha\}.$$

One then verifies that this is a Dedekind cut.

7 Density Versus Completeness

1. State the density property of $\mathbb{Q}$.

Answer. For any rationals $a < b$, there exists rational q with

$$a < q < b.$$

2. Does density mean that every bounded subset of $\mathbb{Q}$ has a supremum in $\mathbb{Q}$?

Answer. No. Density concerns points between two existing points; completeness concerns least upper bounds.

3. Give a bounded subset of $\mathbb{Q}$ with no supremum in $\mathbb{Q}$.

Answer. For example,

$$\{q \in \mathbb{Q} : q^2 < 2, q > 0\}.$$

4. Why does the existence of rationals arbitrarily close to a gap not fill the gap?

Answer. Density produces better approximations but does not produce the missing limiting point itself.

5. Explain the distinction:

there are points arbitrarily close versus the limiting boundary belongs to the space.

Answer. “Arbitrarily close” is an approximation statement. “Belongs to the space” is an existence statement.

6. Can an ordered set be dense but incomplete?

Answer. Yes. $\mathbb{Q}$ is the standard example.

7. Is $\mathbb{Q}$ dense in itself?

Answer. Yes.

8. Is $\mathbb{Q}$ complete?

Answer. No.

9. Why is there no contradiction between the previous two answers?

Answer. Density and completeness are different properties. One does not imply the other.

10. Does the statement “between any two rationals there is another rational” say anything by itself about the existence of suprema?

Answer. No. It says nothing about whether bounded subsets have least upper bounds.

11. In what sense is completeness a global property while density is a local order property?

Answer. Density is local because it concerns pairs of points. Completeness is global because it concerns arbitrary bounded subsets.

12. Why are Dedekind cuts specifically designed to repair incompleteness rather than lack of density?

Answer. $\mathbb{Q}$ already has density. What it lacks are certain least upper bounds, and cuts are designed to add those missing order-locations.

8 Consequences of “No Greatest Element”

1. Write again the quantified form of “ A has no greatest element.”

Answer.

$$\forall a \in A \exists a' \in A \text{ such that } a < a'.$$

2. Prove that a nonempty set with no greatest element must be infinite.

Answer. Every finite nonempty linearly ordered set has a greatest element. Therefore a nonempty set with no greatest element must be infinite.

3. Does no greatest element imply unboundedness above? Give a counterexample if not.

Answer. No. For example,

$$\{q \in \mathbb{Q} : q < 1\}$$

has no greatest element but is bounded above.

4. Starting from any $a_1 \in A$, explain how to construct

$$a_1 < a_2 < a_3 < \cdots$$

with every $a_n \in A$.

Answer. Choose $a_2 > a_1$ in A , then $a_3 > a_2$, and continue recursively.

5. Does the existence of such a strictly increasing sequence imply that it converges?

Answer. No. A strictly increasing sequence may diverge or fail to converge in the ambient space.

6. Does it imply that it is bounded?

Answer. No in general. For example, $a_n = n$ is strictly increasing and unbounded.

7. For a Dedekind lower set, why is every such sequence bounded above?

Answer. Any $b \in B$ is an upper bound for every element of A .

8. Let $a \in A$ and let $\varepsilon > 0$ be rational. Using downward closure, density of $\mathbb{Q}$, and the no-greatest-element property, prove that there exists $a' \in A$ such that

$$a < a' < a + \varepsilon.$$

Answer. Choose $z \in A$ with $a < z$. By density choose rational

$$a < a' < \min(z, a + \varepsilon).$$

Since $a' < z$ and A is downward closed, $a' \in A$.

9. Which hypotheses were actually needed in the previous proof?

Answer. No greatest element, downward closure, and density of $\mathbb{Q}$ were used.

10. Does “no greatest element” alone imply that points of A occur arbitrarily close above every $a \in A$ in an arbitrary ordered set?

Answer. No. In an ordered set with jumps, there may be a next element above a and no points arbitrarily close to it.

11. Why does the structure of $\mathbb{Q}$ matter?

Answer. Density of $\mathbb{Q}$ allows one to refine any strict inequality by inserting another rational.

12. After $\mathbb{R}$ has been constructed, suppose $s = \sup_{\mathbb{R}} A$. Why can s not belong to A ?

Answer. If $s \in A$, then no-greatest-element gives some $a \in A$ with $s < a$, contradicting that s is an upper bound.

13. Is the previous argument available before $\mathbb{R}$ is constructed?

Answer. Not if s is being taken as a real supremum before $\mathbb{R}$ has been constructed.

9 Properties of the Upper Half B

1. Given a lower-set formulation of a cut, define

$$B = \mathbb{Q} \setminus A.$$

Answer.

$$B = \mathbb{Q} \setminus A.$$

2. Prove that B is nonempty.

Answer. Since A is proper, there exists a rational not in A , hence in B .

3. Prove that B is upward closed.

Answer. If $b \in B$ and $c > b$, then $c \notin A$; otherwise downward closure would force $b \in A$. Hence $c \in B$.

4. Prove that every element of B is greater than every element of A .

Answer. If some $b \in B$ satisfied $b \leq a$ for $a \in A$, then downward closure of A would force $b \in A$, contradiction.

5. Prove that every element of B is an upper bound of A .

Answer. Since every $a \in A$ satisfies $a < b$, each $b \in B$ is an upper bound of A .

6. Is every upper bound of A necessarily an element of B ?

Answer. Yes, for rational upper bounds. If $u \in \mathbb{Q}$ is an upper bound of A , then $u \notin A$, because A has no greatest element. Hence $u \in B$.

7. Must B have a least element?

Answer. No.

8. When does B have a least element?

Answer. Exactly when the cut is induced by a rational number.

9. Can B have a greatest element?

Answer. No. Since B is upward closed and nonempty, it contains arbitrarily large rationals.

10. Can B be finite?

Answer. No. The same reasoning shows it is infinite.

11. Suppose $b \in B$ and $b' < b$. Must b' lie in B ?

Answer. Not necessarily. It may lie in A or B .

12. Suppose $b \in B$ and $b' > b$. Must b' lie in B ?

Answer. Yes. Upward closure gives $b' \in B$.

13. Compare the asymmetry:

A has no greatest element

while

B may or may not have a least element.

Why is this asymmetry useful?

Answer. It makes rational cuts identifiable by the existence of $\min B$, while keeping the lower-set convention uniform for all cuts.

10 Approximation Across the Cut

1. Let (A, B) be a Dedekind cut. What would it mean for A and B to be separated by a positive rational gap?

Answer. It would mean that there exists $\delta > 0$ such that

$$b - a \geq \delta$$

for every $a \in A$ and $b \in B$.

2. Formulate precisely the statement

$$\forall \varepsilon \in \mathbb{Q}, \varepsilon > 0, \quad \exists a \in A, \exists b \in B \quad \text{such that} \quad b - a < \varepsilon.$$

Answer.

$$\forall \varepsilon \in \mathbb{Q}_{>0} \exists a \in A \exists b \in B \text{ such that } b - a < \varepsilon.$$

3. Why is this statement stronger than merely saying that every $a \in A$ is less than every $b \in B$?

Answer. Order separation only says $a < b$. The approximation statement says the distance can be made arbitrarily small.

4. Try to prove the approximation statement using only properties of $\mathbb{Q}$ and the cut axioms.

Answer. One standard proof subdivides a rational interval from some $a_0 \in A$ to some $b_0 \in B$ into sufficiently many equal rational pieces and finds adjacent subdivision points lying on opposite sides of the cut.

5. Where does the Archimedean property enter the proof?

Answer. It is used to choose enough subdivisions so that the mesh size is smaller than ε .

6. Which version of the Archimedean property is sufficient?

Answer. It suffices that for every positive rational ε and positive rational M , there exists $n \in \mathbb{N}$ with

$$M/n < \varepsilon.$$

7. Why does invoking that version not require completeness of $\mathbb{R}$?

Answer. This is a property of $\mathbb{Q}$ and $\mathbb{N}$ and can be proved without real completeness.

8. Could the approximation result be proved directly from density of $\mathbb{Q}$ alone?

Answer. Not by density alone in the most direct argument. Density inserts points between known rationals but does not by itself give a uniform finite subdivision argument.

9. If there were a fixed rational $\delta > 0$ such that

$$b - a \geq \delta$$

for every $a \in A$ and $b \in B$, what would go wrong?

Answer. Such a gap would allow rational points between the two halves that belong to neither side, contradicting that $A \cup B = \mathbb{Q}$, once sufficiently fine rational subdivisions are considered.

10. How does the no-greatest-element property help rule out such a gap?

Answer. If $a \in A$, there are larger elements of A . Thus A cannot terminate a positive distance before all of B .

11. Does the approximation property imply that there exists a rational number lying exactly “between” A and B ?

Answer. No. Arbitrarily small gaps do not imply that a rational realizes the boundary.

12. Why is the difference between “arbitrarily close” and “attained boundary” important here?

Answer. A sequence of rational approximations may approach a missing location without any rational being equal to that location.

11 Arithmetic on Dedekind Cuts

1. If real numbers are Dedekind cuts, what should it mean to add two real numbers?

Answer. One defines an operation on cuts that corresponds to rational addition and then proves that the result is again a cut.

2. Given cuts A and C , suggest a definition for their sum using rational sums $a + c$.

Answer. A natural definition is

$$A + C = \{q \in \mathbb{Q} : \exists a \in A, \exists c \in C, q < a + c\},$$

or an equivalent downward-closed formulation.

3. Why is it not enough merely to define a set? What must be proved about the resulting set?

Answer. Because a real number is a cut, the output must satisfy all Dedekind-cut axioms.

4. Which cut axioms must be checked for addition?

Answer. Nonemptiness, properness, downward closure, and absence of a greatest element.

5. Why is multiplication more complicated than addition?

Answer. Products interact with order differently depending on signs.

6. Why do signs create complications when defining multiplication of cuts?

Answer. Multiplication by a negative number reverses inequalities, so a single naive formula does not work uniformly.

7. What should the additive identity look like as a cut?

Answer. The additive identity is the cut

$$\{q \in \mathbb{Q} : q < 0\}.$$

8. What should the cut representing a rational number r look like?

Answer.

$$A_r = \{q \in \mathbb{Q} : q < r\}.$$

9. What should negation of a cut accomplish order-theoretically?

Answer. It should produce a cut whose represented value is the order-theoretic reflection of the original one about 0.

10. Why is defining the reciprocal of a positive cut more delicate than defining its sum with another cut?

Answer. Reciprocals reverse inequalities on positive numbers and require care near 0.

11. After defining arithmetic, what must be proved before one may call the resulting structure an ordered field?

Answer. One must verify the field axioms, compatibility with the order, and that the rational embedding preserves the arithmetic.

12 Completeness After the Construction

1. What is the logical order of the Dedekind construction? Arrange the following:

$\mathbb{Q}$, Dedekind cuts, $\mathbb{R}$, least-upper-bound property.

Answer.

$\mathbb{Q} \longrightarrow \text{Dedekind cuts} \longrightarrow \mathbb{R} \longrightarrow \text{proof of the least-upper-bound property}.$

2. Why must completeness be proved after the real numbers have been defined?

Answer. Completeness is a theorem about the newly defined structure, not a premise that may be assumed beforehand.

3. Suppose $\mathcal{S}$ is a nonempty set of Dedekind cuts bounded above. What kind of object should $\sup \mathcal{S}$ itself be?

Answer. It should itself be a Dedekind cut, hence an element of the constructed $\mathbb{R}$.

4. A common construction defines the supremum of a family of cuts by a union. Why is

$$\bigcup_{A \in \mathcal{S}} A$$

a natural candidate?

Answer. The union collects every rational lying below at least one cut in the family, which is exactly what a least upper bound should do.

5. Which Dedekind-cut axioms must be verified for that union?

Answer. One must verify nonemptiness, properness, downward closure, and no greatest element.

6. Why does this proof of completeness avoid circular reasoning?

Answer. The candidate supremum is constructed explicitly from sets of rationals, without invoking a previously existing supremum in $\mathbb{R}$.

7. What is the conceptual difference between:

assuming every bounded set has a supremum

and

constructing the supremum explicitly as another cut?

Answer. The first is an existence axiom or theorem. The second provides a concrete object and proves that it has the required property.

8. Once completeness of the cut construction has been proved, why is it thereafter legitimate to write

$$\sup_{\mathbb{R}} S$$

for suitable bounded sets $S \subseteq \mathbb{R}$?

Answer. Because completeness has now been established as a theorem of the constructed real number system.

9. At that later stage, how can one reinterpret a Dedekind cut geometrically as

$$A = \{q \in \mathbb{Q} : q < \alpha\}$$

for some $\alpha \in \mathbb{R}$?

Answer. For $\alpha \in \mathbb{R}$, its corresponding lower rational set is

$$\{q \in \mathbb{Q} : q < \alpha\}.$$

10. Why would using that geometric representation at the beginning of the construction reverse the logical dependency?

Answer. It would assume the real number α exists before the construction that is supposed to define it.

13 Mixed Assumption Checks

For each statement, decide whether it is true, false, or meaningless without further specification. If it is ambiguous, identify the hidden assumption.

1. Every Dedekind cut has a supremum.

Answer. Ambiguous. It depends on the ambient ordered set. In $\mathbb{R}$, yes; in $\mathbb{Q}$, not always.

2. Every Dedekind lower set is bounded above.

Answer. True. Any $b \in B$ is a rational upper bound of A .

3. Every Dedekind lower set has no maximum.

Answer. True. This is part of the standard definition.

4. Every Dedekind upper set has a minimum.

Answer. False. It has a minimum exactly for rational cuts under the usual convention.

5. If A has no greatest element, then A is unbounded above.

Answer. False.

6. If A is bounded above, then $\sup A$ exists.

Answer. False in general. It requires completeness of the ambient ordered set.

7. If A is a Dedekind lower set, then $\sup_{\mathbb{Q}} A$ exists.

Answer. False.

8. If A is a Dedekind lower set, then $\sup_{\mathbb{R}} A$ exists.

Answer. True after $\mathbb{R}$ has been constructed and proved complete.

9. Every cut has a boundary point.

Answer. Ambiguous. Geometrically yes after embedding into $\mathbb{R}$; not necessarily as a rational point.

10. Every cut has a rational boundary point.

Answer. False.

11. Every cut corresponds to exactly one real number.

Answer. True after the Dedekind construction, because cuts are the real numbers in that model.

12. Every real number is first defined independently and then represented by a cut.

Answer. False. In the construction, the cut itself defines the real number.

13. The density of $\mathbb{Q}$ implies that $\mathbb{Q}$ has no gaps.

Answer. False. $\mathbb{Q}$ is dense but incomplete.

14. A set can be bounded above and still fail to have a supremum in its ambient ordered set.

Answer. True.

15. A set can have no greatest element and still have a supremum.

Answer. True.

16. If B has a least element, the cut corresponds to a rational number.

Answer. True.

17. If B has no least element, the cut corresponds to a new real number not represented by a rational.

Answer. True, meaning a cut not induced by any rational.

18. The proof that cuts approximate each other arbitrarily closely must use completeness of $\mathbb{R}$.

Answer. False.

19. The Archimedean property cannot be used before $\mathbb{R}$ is constructed.

Answer. False. Appropriate Archimedean properties can be proved already in $\mathbb{Q}$.

20. Once $\mathbb{R}$ is constructed and proved complete, one may freely reinterpret cuts using real suprema.

Answer. True.

14 Dependency Questions

For each proposed argument, identify whether it is logically valid at the stage where real numbers are being constructed from rational Dedekind cuts.

1. “ A is bounded above, therefore let $s = \sup A$.”

Answer. Invalid without specifying a complete ambient ordered set. Boundedness alone does not imply existence of a supremum in $\mathbb{Q}$.

2. “Choose $b \in B$. Then b is an upper bound of A .”

Answer. Valid. The order-separation property of a cut makes every $b \in B$ an upper bound of A .

3. “Since $\mathbb{Q}$ is dense, choose a rational number strictly between two distinct rationals.”

Answer. Valid. Density of $\mathbb{Q}$ is available before constructing $\mathbb{R}$.

4. “The cut representing $\sqrt{2}$ consists of all rationals below the already existing real number $\sqrt{2}$.”

Answer. Invalid as a foundational definition. It presupposes the real number $\sqrt{2}$.

5. “Define the object called $\sqrt{2}$ to be the cut

$$\{q \in \mathbb{Q} : q < 0 \text{ or } q^2 < 2\}.$$

”

Answer. Valid, provided the displayed set is proved to satisfy the cut axioms.

6. “Because $\mathbb{R}$ is complete, the cut has a least upper bound.”

Answer. Invalid during the construction. It assumes the completeness of the structure being constructed.

7. “Because the Archimedean property holds in $\mathbb{Q}$, choose $n \in \mathbb{N}$ such that $n\varepsilon$ exceeds a prescribed rational number.”

Answer. Valid if that Archimedean statement has already been proved for $\mathbb{Q}$.

8. “Because A has no greatest element, choose $a' \in A$ with $a < a'$.”

Answer. Valid. This is exactly the no-greatest-element axiom.

9. “Because B is nonempty, choose an upper bound of A .”

Answer. Valid. Any $b \in B$ is an upper bound of A .

10. “Because every cut is itself defined to be a real number, no prior real boundary is needed.”

Answer. Valid. This is the conceptual basis of the construction.